Hareshkumar Jadav

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Education

PhD **Indian Institute of Technology Indore**, Computer Science — Jul 2023 – Present

- CGPA: 9.24/10
- Supervisor: Dr. Ranveer Singh, IIT Indore, India
- Co-Supervisor: Dr. Sudipta Mallik, Marshall University, USA

M.Sc **The Maharaja Sayajirao University of Baroda**, Mathematical Sciences — Aug 2020 – May 2022

- CGPA: 8.1/10

B.Sc **Dr. Subhash Science College Junagadh** , Mathematical Sciences — Jul 2017 – Apr 2020

- CGPA: 7.9/10

Research Interest

Algorithm, Algebraic Graph Theory, Linear Algebra, Combinatorial optimization

Programming skills:

C/C++, Python, LaTeX, Matlab, SageMath

Experience

Visiting Scholar — Jul 2025 - Oct 2025

- School of Industrial Engineering, Purdue University, USA
  Project Supervisor: Prof. Vaneet Aggarwal
  Project title: Stronger Approximation Guarantees for Non-Monotone γ-Weakly DR-Submodular Maximization

Junior Research Fellow — Nov 2022 - Jul 2023

- Computer Science and Engineering Department, IIT Indore
  Project Supervisor: Dr. Ranveer Singh
  Project title: The Construction of Expander and Ramanujan Graphs and Their Application in Robust Network topologies

Lecturer — Sept 2022 - Oct 2022

- Department of Mathematics
  Dr. Subhash University Junagadh, Gujarat

Publications

Upper-Linearizability of Online Non-Monotone DR-Submodular Maximization over Down-Closed Convex Sets — Feb 2026

Yiyang Lu, ***Hareshkumar Jadav***, Mohammad Pedramfar, Ranveer Singh, Vaneet Aggarwal

arXiv preprint, link

Stronger Approximation Guarantees for Non-Monotone γ-Weakly DR-Submodular Maximization — May 2026

Hareshkumar Jadav, Ranveer Singh, Vaneet Aggarwal

AAMAS 2026, link

A construction of random bigraphs and their application to error correction codes Jan 2025
Hareshkumar Jadav, Ashita Gupta, Ranveer Singh
Journal of Statistical Mechanics: Theory and Experiment, link

Strengthening Wilf's inequality on clique number Jul 2025
Hareshkumar Jadav, Sreekara Madyastha, Rahul Raut, Ranveer Singh
ISSAC 2025, link

Cartesian Prime Graphs and Cospectral Families Jul 2025
Abhinav Bitragunta,***Hareshkumar Jadav***, Ranveer Singh
ISSAC 2025, link

Generating Constrained Lattice Paths in a Grid related to counting cycles Sept 2025
Hareshkumar Jadav,Mihir Patel, Samip Shah, Ranveer Singh, Harsh Talati
ICTCS 2025, link

Book drawing algorithm for a fixed vertex order Under review
Geetanath Gonam, ***Hareshkumar Jadav***, Dhananjay Kansule, Ranveer Singh

Moore–Penrose Inverse of Incidence Matrices of Unicyclic Graphs Under review
Hareshkumar Jadav, Priyanshu Pant, Ranveer Singh, Virendra Singh

Moore–Penrose Inverses of Laplacians of Unicyclic Graphs Under review
Hareshkumar Jadav, Priyanshu Pant, Ranveer Singh, Virendra Singh

Stronger Spectral Bound for Clique Number Under review
Hareshkumar Jadav, Sudipta Mallik, Akhilesh Panchangam, Ranveer Singh

Achievements & Awards

- Selected for short-term collaborative research fellowship 2024-25 to visit Prof. Vaneet Aggarwal's CLAN Labs for three months at Purdue University, USA, Funded by IIT Indore.
- Received Travel Support from IIT Indore to attend the 26th Conference of the International Linear Algebra Society, held from June 23–27, 2025, in Kaohsiung.
- Qualified GATE (Computer Science) in February 2023.
- Qualified CSIR-NET (Mathematical Science) in September 2022 with All India Rank-61.
- Qualified CSIR-NET (Mathematical Science) in February 2022 with All India Rank-47.

Academic Responsibilities

- Member, Events & Outreach Team, Department of Computer Science and Engineering, IIT Indore, 2024.
- Member, Organizing Team, Workshop on *Ramanujan Graphs*, held at IIT Indore, October 16, 2024.
- Member, Organizing Team, High-End Workshop on *Mathematical Foundation of Machine Learning and Its Application to Emerging Areas of AI*, held at IIT Indore, May 27–June 1, 2024.
- Member, Organizing Team, Advance Instructional School on Algorithmic Graph Theory, held at IIT Indore, June 26–July 15, 2023.

Workshop & Conference

- Presented a poster at the 40th Annual Conference of the Ramanujan Mathematical Society, held at IIIT Delhi, December 18–20, 2025.
- Attended the 26th Conference of the International Linear Algebra Society, held in Kaohsiung, June 23–27, 2025.
- Attended the Summer School on Approximation Algorithms 2025, held at IISc Bangalore, May 21–25, 2025.
- Attended ACM India Summer School on Algorithmic Approaches to NP-Completeness, held at PSG College of Technology, Coimbatore, July 15–26, 2024.
- Attended Mathematical Foundation of Machine Learning and its Application to the Emerging Area of AI, held at IIT Indore, May 27–June 1, 2024.
- Attended Spectral Method in Algorithms, FSTTCS 2023 Workshop, held at IIIT Hyderabad, December 21–22, 2023.
- Attended Advance Instructional School on Algorithmic Graph Theory, held at IIT Indore, June 26–July 15, 2023.
- Attended Advance Instructional School on Combinatorics and Graph Theory, held at the University of Kashmir, May 1–13, 2023.
- Attended CREST School: Applied Combinatorics, Cryptography and Combinatorial Topology, held at TCG CREST, Kolkata, July 11–22, 2022.
- Attended the State Level Workshop on Problem Solving and Enrichment in Mathematics, December 24–27, 2019.

References

- **Dr. Ranveer Singh**
  Associate Professor, Department of Computer Science, IIT Indore, India
  Email: ranveer@iiti.ac.in
- **Dr. Sudipta Mallik**
  Assistant Professor, Department of Mathematics and Physics, Marshall University, USA
  Email: mallik@marshall.edu
- **Prof. Vaneet Aggarwal**
  Professor, School of Industrial Engineering Purdue University, USA
  Email: vaneet@purdue.edu